

Lesson 3 worksheet

Lesson 03

1. The functions f and fg are defined by $f(x) = e^{(x^2 + 3)}$ for $x < 0$ and $fg(x) = e^{(2x)}$ for $x > 3/2$. Explain why f^{-1} exists. [1 mark]

2. Hence find and simplify an expression for $g(x)$. [2 marks]

3. Find the possible values of the constant k for which $|(x^2 - 4)(x - 2)| = k$ has exactly 4 different solutions. [2 marks]

4. Sketch the graph of $y = |(x + 3)(x - 4)|$, stating the intercepts with the axes [2 marks]

5. A function g is such that $g(x) = x^{(1/2)} + 4$, $x \geq 0$. Find the exact solution of the equation $g(f(x)) = 12$ [4 marks]

6. A function h is defined by $h(x) = 1/x^2 + 2$ for $x > 0$. Find an expression for $gh(x)$ in simplest form. [2 marks]

7. Write down the range of g^{-1} . [1 mark]

8. In the case where $a=5$, solve the equation $gf(x)=3$. Give your answer correct to 3 decimal places. [3 marks]

9. It is given that $f(x)=2e^x+a$ for $x\geq 0$, where a is an integer and $g(x)=\sqrt{x-1}$ for $x\geq 1$. Find the least value of a so that the function gf exists for all $x\geq 0$. [2 marks]
